

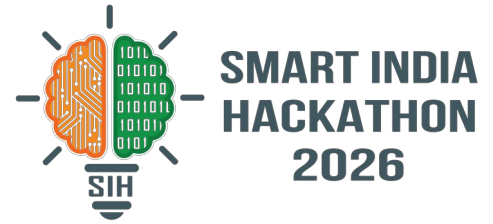
SMART INDIA HACKATHON 2026

MailTrace

Explainable email threat detection with origin and campaign forensics

Upload a saved email. See the evidence behind its risk and likely infrastructure.

PROBLEM STATEMENT ID	SIH26106
PROBLEM STATEMENT TITLE	AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	[ENTER PORTAL TEAM ID]
TEAM NAME	[ENTER REGISTERED TEAM NAME]



From one suspicious message to a connected case file.

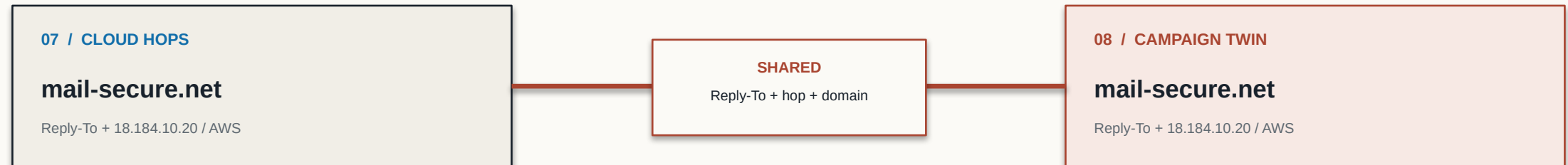
MailTrace combines detection, origin context and campaign correlation in one analyst workflow.

PROPOSED SOLUTION



INNOVATION PROOF

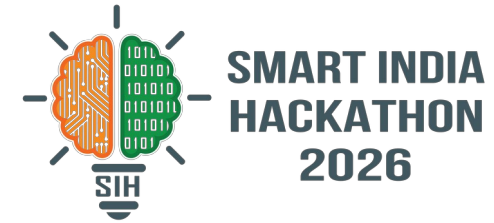
CAMPAIGN CORRELATION / TWO CASES, ONE EXPLAINABLE RELATIONSHIP



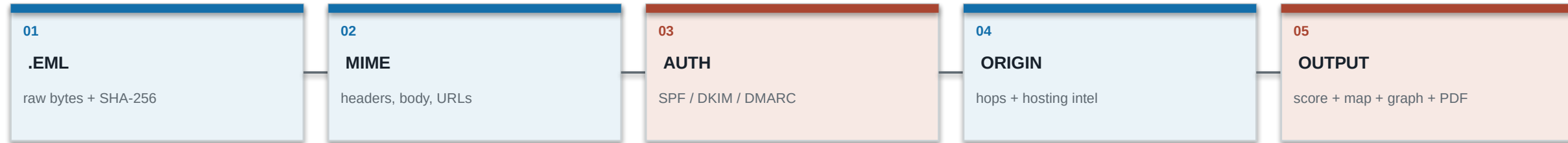
Correlation creates a campaign candidate — never an identity or exact-location claim.

Evidence pipeline: small, explainable modules.

The working prototype is deterministic first; NLP is one supporting signal, not the whole product.



METHODOLOGY



IMPLEMENTED PROTOTYPE

- FastAPI + Python email parser + SQLite case store
- NetworkX campaign graph + Leaflet hop map
- ReportLab forensic PDF with exact-file SHA-256
- Offline demo intelligence keeps laptop runs reproducible

REAL LOCALHOST PROOF



Captured from the local case screen: score, reasons, hop map and evidence footer.

Feasible now; harden the edges as adoption grows.

The first version runs on a laptop and avoids fragile live dependencies. The production path is modular.

FEASIBILITY

NOW / IDEA-ROUND PROTOTYPE

- Saved .eml upload; no mailbox credentials
- Eight crafted cases for spoof, phish, BEC and campaign paths
- Offline demo geo/domain cache for reproducible runs
- Local case store, graph, map and forensic PDF

NEXT / PRODUCTION HARDENING

- Live DNS and cryptographic auth verification
- Maintained GeoIP/ASN data and private deployment
- Access control, PII masking and retention policies
- Larger labelled corpus, calibration and analyst feedback

RISKS AND MITIGATIONS

RISK

MITIGATION

Headers may be forged

Show confidence; preserve raw evidence; never call a lead proof.

IP geo is not a person

Report hosting city/ISP only; explicitly reject GPS attribution.

Email is sensitive

Local-first workflow; no Gmail ingest; future masking and retention controls.

Impact: turn a vague warning into an actionable case.

MailTrace helps people explain what happened, preserve evidence and decide what to do next.

ANALYST DIFFERENCE

FILTER

spam / not spam

A label leaves the next question unanswered.

MAILTRACE CASE FILE

SPF fail · Reply-To mismatch · earliest hop

Frankfurt / AWS · related case 08 · SHA-256

TARGET USERS AND VALUE

IT / helpdesk	Triage a suspicious message with reasons they can cite.
SOC / incident response	Preserve original evidence and connect related messages.
Universities / SMEs	Run locally without handing an inbox to a third party.

PILOT MEASURES / NO INVENTED NUMBERS

time-to-triage · evidence completeness · linked-campaign discovery · analyst agreement

Research basis and implementation references.

The prototype maps directly to SIH26106 while keeping attribution and privacy claims bounded.

PROBLEM + STANDARDS

SIH26106	Official PS: NLP/ML detection, header forensics, IP geo, domain intelligence, graph correlation and forensic reporting.
RFC 5322	Internet Message Format: structured headers, message IDs and Received fields.
RFC 7208 / 6376 / 7489	SPF, DKIM and DMARC concepts used for authentication evidence and alignment.

BUILD REFERENCES

Python email	MIME parsing; attachments named and hashed, never executed.
NetworkX + Leaflet	Relationship graph and hop map for analyst inspection.
ReportLab + SHA-256	Portable case PDF and integrity fingerprint of the exact uploaded bytes.

SCOPE AND SAFETY

- Local crafted .eml files only; no Gmail/Outlook ingest and no real phishing messages.
- Geo output is hosting infrastructure with confidence, never the exact location of a person.
- Evidence hash is an integrity fingerprint, not a public blockchain or cryptocurrency ledger.

Source links: sih.gov.in/sih2026PS · rfc-editor.org · docs.python.org · networkx.org · leafletjs.com